SCHOOLOF ENGINEERING

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

B.TECH DEGREE II SEMESTER SECOND INTERNAL EXAMINATION, JULY 2024

Sem: S2 EC/ IT A&B

Course Title: 23-200-0203B : DIGITAL ELECTRONICS (2023 Scheme)

Faculty: Dr Deepa Sankar, Dr Binu Paul

Time: 2Hrs **Max. Marks: 40**

Course Outcomes On successful completion of teaching-learning and valuation activities, a student would be able to

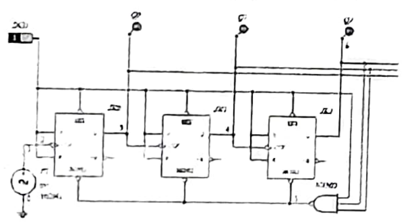
CO1 Understand Boolean postulates and Basic building blocks of Digital systems

CO2 Design Optimal digital circuits using basic building blocks

CO3 Analyse basic digital circuits

CO4 Understand HDL models of simple circuits

L1:Remember, L2:Understand ,L3:Apply L4: Analyse L5: Evaluate L6: Create

Qn no	***PART A (Answer ALL questions: marks 4 x 1= 4)***	**CO ,PO,BL**	
1	For a 5 bit data , Hamming code size is (a)6 (b) 7 (c) 8 (d) 9	**1,(123)1**	
2	Race around condition exist in ___ flip flop (a) **D** (b) **JK** (c) **Master Slave** (d) **A.B '**	**1,(123)1**	
3	Identify the modulus of the counter below.  (a)3 (b) 6 (c) 4 (d) 6	**3,(1234)4**	
4	Fanout of a TTL gate with following spec is $I_{OL}(max) = 32$ mA, $I_{IL}(max) = 1.6$ mA, $I_{OH}(max) = 400$ uA, $I_{IH}(max) = 10$ uA (a)10 (b) 20 (c) 30 (d) 40	**3,(123)1**	2
Qn no	***PART B (Answer Any THREE: marks 3x2= 6)***	**CO ,PO,BL**	
	What is the purpose of a Hamming code? Give the Hamming code for the 4 bit data sequence 1011. Draw the circuit to generate the Hamming code for the above case. Explain how error can be corrected, with help of a suitable example.	**1,(123)2**	12
6	What are shift registers? Explain various types with proper circuit diagrams. Give one application for each.	**1,(123)2**	
7	Explain the architecture of a PLA? Schematically show the layout of a Full adder on a PLA of dimensions 3x8x4.	**2,(1234)3**	12
8	What is meant by Figure of Merit of gate? Explain. Given the following specifications evaluate Figure of merit of the component. Operating Voltage = 5.1V, Current drawn under High and Low conditions = 20μA , 10 mA. $t_{p\,LH} = 5$ ns $t_{pHL}=15$ ns	**2,(123)4**	12
9	Draw and explain the working of a serial adder. What are its advantages over a parallel adder.	**2,(123)3**	
10	Give the excitation table, state diagram and characteristic equation of a JK flip flop	**2,(123)3**	12

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Qn no	***PART C (Answer Any THREE: marks 3 x 10=30)***	**CO ,PO,BL**
11	What is meant by a Carry Look ahead adder? Draw the circuit for a 4 bit case by deriving relevant Boolean expressions. Quantitively Explain the advantage it offers over a ripple carry adder.	**1,(123)4**
12	Design a mod 4 synchronous counter. Draw the circuit using JK flip flops configured as T flip flop. With necessary combinational circuit produce 3 o/ps whose periods are in the ratio 1:1:2, from the above counter output	**2,(123)3**
13	With neat circuit diagram, explain the working of the 2 i/p NAND gate belonging to TTL family with Totem-pole output stage. What is the role of diode in the output stage.	**2,(123)2**
14	Design Modulo 5 synchronous counter counting as 1-2-3-7-6-1-2-3-7-6.... (without Lock out)	**2,(123)4**
15	Design Modulo 5 counters, both synchronous and Asynchronous giving proper steps and explanation.	**2,(123)2**
16	Design a shift register based sequence generator to give the following ...10110...	**2,(123)3**
17	Explain the typical current and voltage parameters (V_{IH}, V_{OH}, V_{OL}, V_{IL}, I_{IH}, I_{OH}, I_{OL}, I_{IL}) of a logic family. Explain significance of Noise margin, Fan out .	**2,(123)2**

	1	2	3	4
CO%	18.5%	79%	2.5%	0%
BL%	3.5%	39.5%	30%	27%